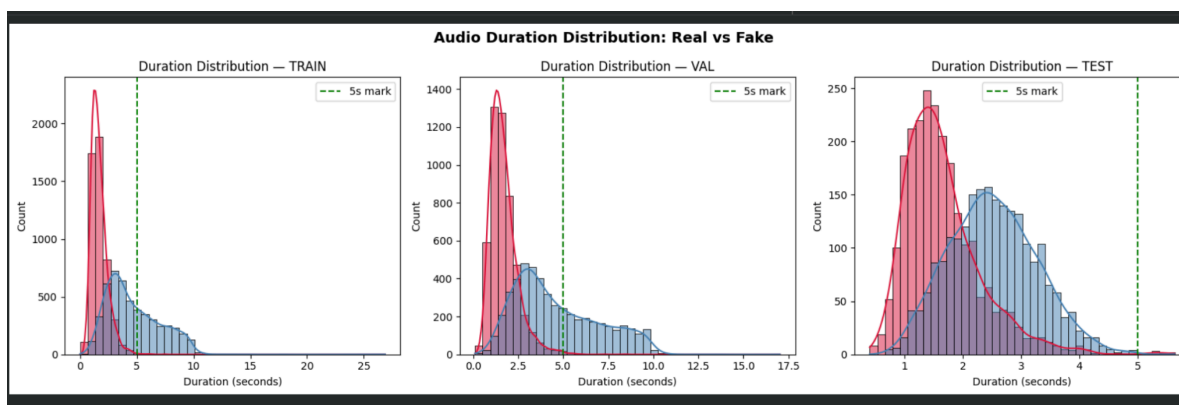
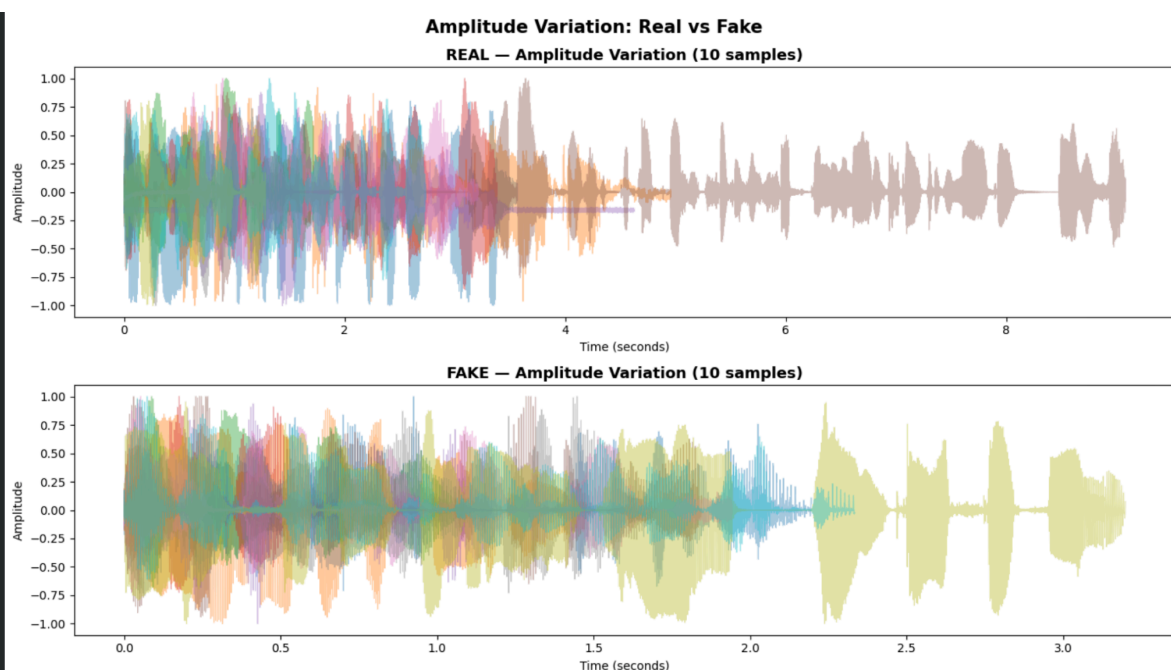
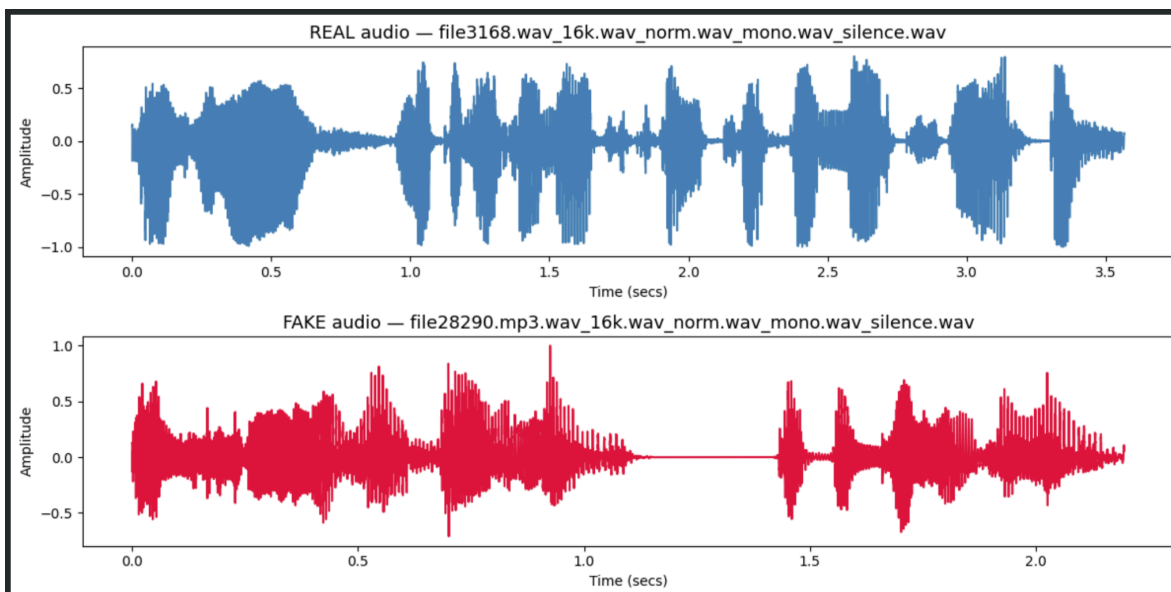




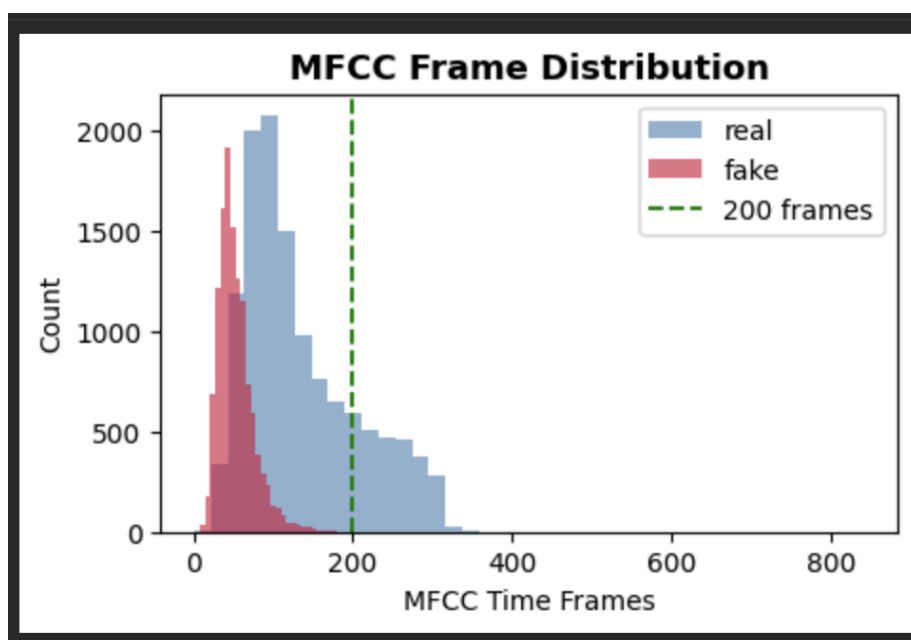
- Train → real: 26941 files
Train → fake: 26927 files

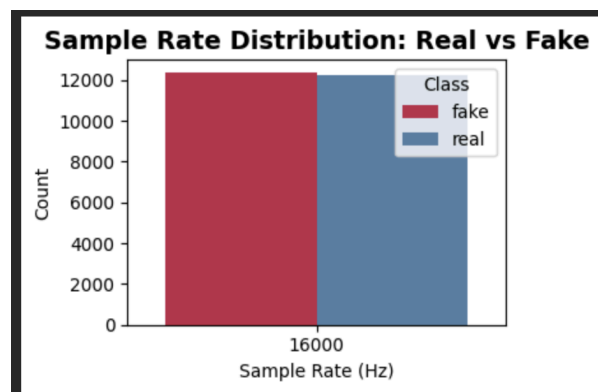
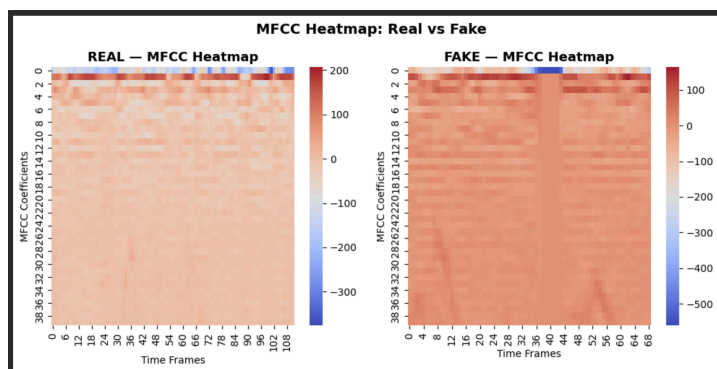
Val → real: 5400 files
Val → fake: 5398 files

Test → real: 2264 files
Test → fake: 2370 files

```
train/real: 100%|██████████| 5000/5000 [01:03<00:00, 78.81it/s]
train/fake: 100%|██████████| 5000/5000 [00:48<00:00, 103.37it/s]
test/real: 100%|██████████| 2264/2264 [00:19<00:00, 114.99it/s]
test/fake: 100%|██████████| 2370/2370 [00:18<00:00, 131.12it/s]
val/real: 100%|██████████| 5000/5000 [00:55<00:00, 90.34it/s]
val/fake: 100%|██████████| 5000/5000 [00:37<00:00, 133.94it/s]
/n EDA collected on 24634 samples
```

	split	class	sample_rate	duration	mfcc_frames
0	train	real	16000	4.874313	153
1	train	real	16000	8.695312	272
2	train	real	16000	2.022500	64
3	train	real	16000	4.987563	156
4	train	real	16000	3.124375	98





Processing training data...

real: 100%|██████████| 26941/26941 [16:29<00:00, 27.24it/s]

fake: 100%|██████████| 26927/26927 [15:53<00:00, 28.23it/s]

X_train: (53866, 40, 156, 1), y_train: (53866,)

Processing validation data...

real: 100%|██████████| 5400/5400 [01:08<00:00, 79.05it/s]

fake: 100%|██████████| 5398/5398 [01:01<00:00, 87.65it/s]

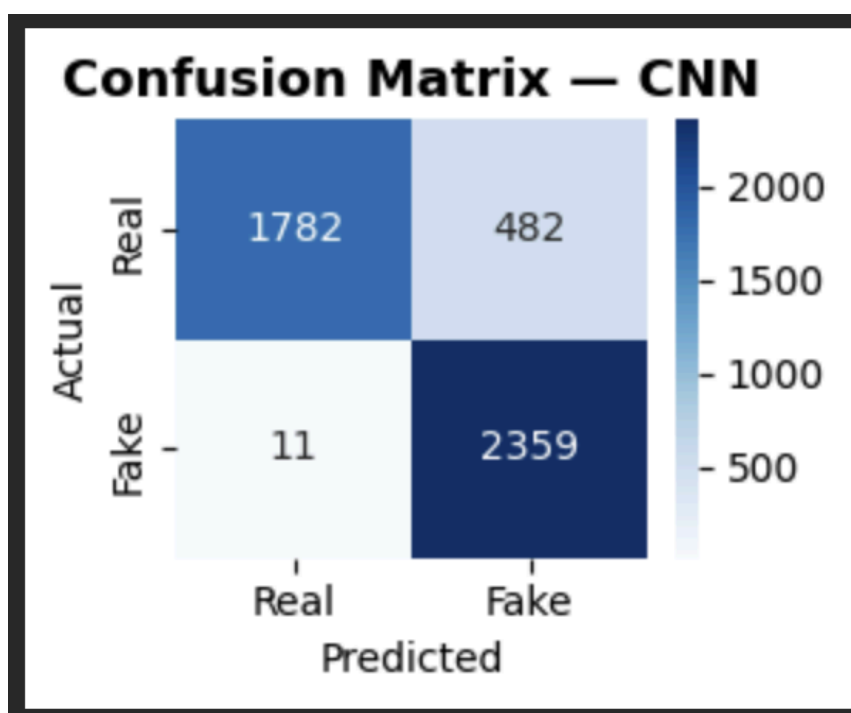
X_val: (10798, 40, 156, 1), y_val: (10798,)

Processing test data...

real: 100%|██████████| 2264/2264 [00:26<00:00, 84.41it/s]

fake: 100%|██████████| 2370/2370 [00:26<00:00, 88.68it/s]

X_test: (4634, 40, 156, 1), y_test: (4634,)



```
... Per-Class Accuracy:
    Real (Genuine) : 78.71% (target ≥ 75%)
    Fake (Deepfake): 99.54% (target ≥ 75%)
```

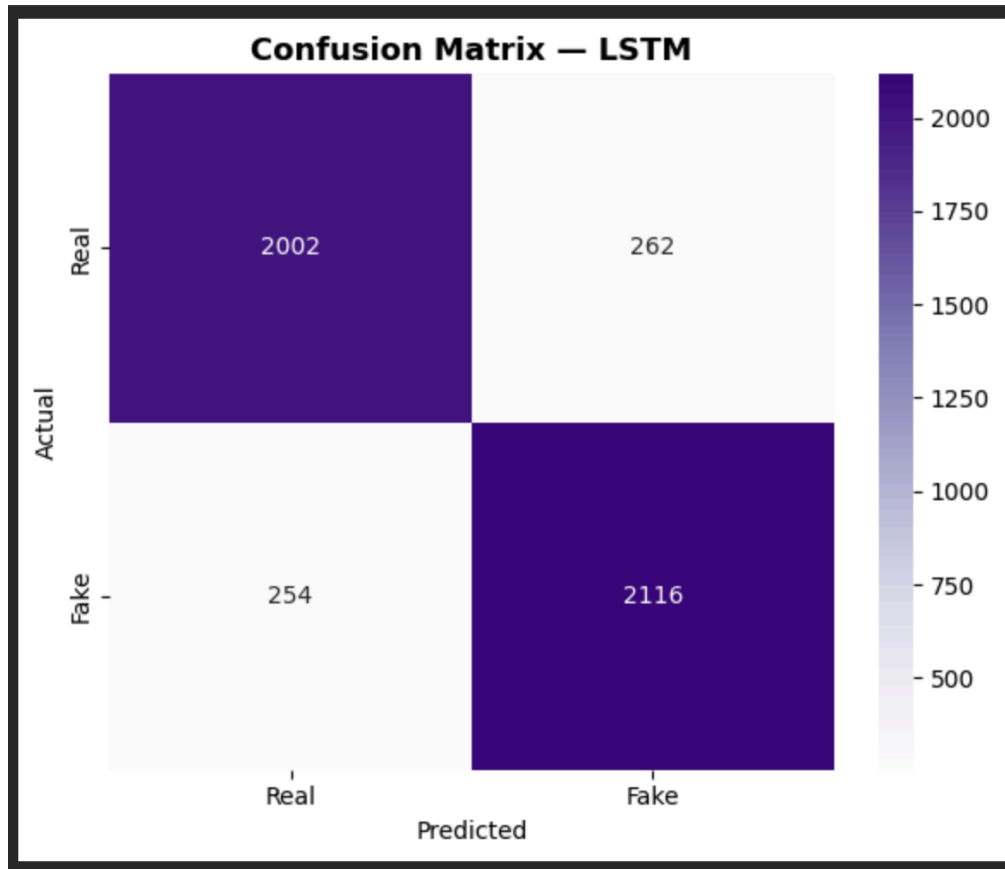
```
=====
FINAL CNN EVALUATION REPORT
=====
```

```
✓ Test Accuracy      : 89.36% (target ≥ 80%)
✓ F1 Score           : 90.54% (target ≥ 80%)
✓ EER                : 3.26% (target ≤ 12%)
✓ Real Accuracy      : 78.71% (target ≥ 75%)
✓ Fake Accuracy      : 99.54% (target ≥ 75%)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 156, 128)	86,528
dropout (Dropout)	(None, 156, 128)	0
lstm_1 (LSTM)	(None, 64)	49,408
dropout_1 (Dropout)	(None, 64)	0
dense (Dense)	(None, 64)	4,160
batch_normalization (BatchNormalization)	(None, 64)	256
dropout_2 (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 1)	65

Total params: 140,417 (548.50 KB)
Trainable params: 140,289 (548.00 KB)
Non-trainable params: 128 (512.00 B)



```
=====
FINAL LSTM EVALUATION REPORT
=====
```

✓	Test Accuracy	: 88.86%	(target ≥ 80%)
✓	F1 Score	: 89.13%	(target ≥ 80%)
✓	EER	: 11.16%	(target ≤ 12%)
✓	Real Accuracy	: 88.43%	(target ≥ 75%)
✓	Fake Accuracy	: 89.28%	(target ≥ 75%)

```
=====
CNN vs LSTM — FINAL COMPARISON
=====
```

Metric	CNN	LSTM
Test Accuracy	92.73%	88.86%
F1 Score	93.30%	89.13%
EER	4.38%	11.16%
Real Accuracy	86.17%	88.43%
Fake Accuracy	98.99%	89.28%

✓ Selected Model: CNN
Reason: Higher accuracy, F1, and significantly better EER (4.38% vs 11.16%)

